

IPBMA - Practical case 1: Methodology for determining the level of redness from bulbar conjunctiva images

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I. INTRODUCTION

The main goal of this practice is to implement a methodology to determine the level of redness of the eye using images of the bulbar conjunctiva. The image set provided consists of 10 images in two different formats, "jpg" and "BMP", the dimensions of the images are the same (1024x768) and there is a combination in terms of the gaze direction of the patient, this means, in some photos the pupil or the eye is looking toward the left and in others toward the right.

II. METHODOLOGY

Several approaches and ways were tested, however, this part is going to be explained the process and the chosen techniques for getting the final classification. The discarded procedures will be explained in the next section.

The methodology was divided into 4 tasks that are explored in the following sections:

A. Region of interest (ROI)

The first step was to identify the region of interest of the eye to be analysed. After several cropping trials, finally, it was decided to select the area within the next ranges: y-axis between 270 and 550, and x-axis between 300:750. An example of the cropping is shown in Figure 1.

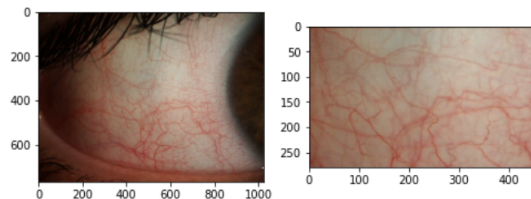


Fig. 1. Example of traffic signs.

B. Location of blood vessels within the ROI

Once the region of interest was selected, several methods were applied for locating the blood vessels. In Figure 2, the output of each step is shown: cropped image, colour space and channel selected, CLAHE and binary segmentation.

1) *Color space selection*: First of all, the colour space was analysed in order to decide in which one of the blood vessels of the bulbar conjunctiva was more clear with respect to the white part. After looking into some of colour spaces, the Green channel from the RGB colour space was selected.

2) *Contrast Limited Adaptive Histogram Equalization (CLAHE)*: For obtaining an image with higher global contrast, a histogram equalization (HE) is recommended. Actually, for further histogram equalization improvements, an algorithm called Contrast Limited Adaptive Histogram Equalization (CLAHE) it might be applied. Contrast Limited Adaptive Histogram Equalization (CLAHE) is a variant of Adaptive Histogram Equalization, where the adaptive method computes several histograms, each corresponding to a distinct section of the image, and uses them to redistribute the lightness values of the image. However, AHE has a tendency to overamplify noise in relatively homogeneous regions of an image, CLAHE prevents this by limiting the amplification. So, a CLAHE was applied where the parameters used for the method were: 5.0 as a threshold for contrast limiting and (8,8) as the size of the tiles for applying the histogram equalization.

3) *Thresholding*: After the equalization of the image, it was sought to convert the grayscale image to a binary image, in order to segment the foreground, vessels in this case, and background of the input image. For that, a global thresholding technique was used for this segmentation using a threshold value of 100 and a Binary method.

C. Occupancy of blood vessels

For estimating the occupancy of the blood vessels in relation to the total of the eye region, basically, black pixels (values equal to zero) were gotten and divided by the total of pixels within the ROI.

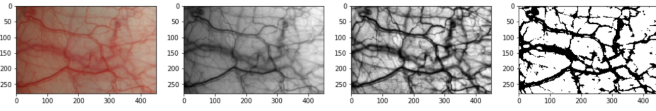


Fig. 2. Cropped image, Green channel (RGB), CLAHE and binary segmentation

D. Classification

Finally, for classifying, looking at the values or proportion of vessels gotten in the previous step, two thresholds were defined for our case in particular. With this, the classification might be shown in Table 3 and its confusion matrix in Figure ?? . The results gotten with this specific dataset are good in terms of there being only one case where the methodology fails. However, it would be interesting to test the proposed approach on other images to check its stability of it.

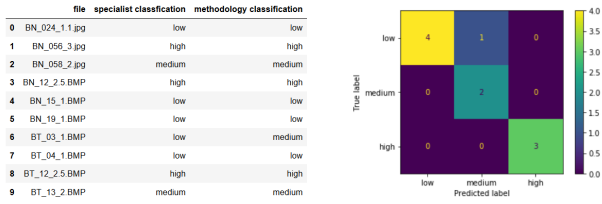


Fig. 3. Specialist vs methodology classification; Confusion matrix

III. DISCARDED TECHNIQUES

A. Cropping using an ellipse mask

The type of cropping shown in Figure 4 was thought in order to remove some tips of the eyelashes that were still visible, for some of the images, in the selected ROI defined above. The main issue was that this solution also hid some vessels from the corners, and using it, some information was lost for the calculation.

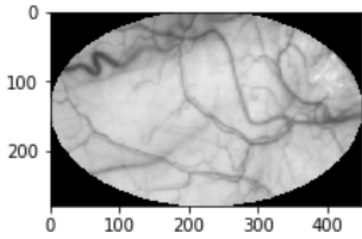


Fig. 4. Cropping using an ellipse mask

B. Removing black parts

The initial idea was to remove black or very dark parts of the eye such as eyelashes and the pupil, for selecting a bigger region of the eye and being able to discard them in the binarization part. However, after trying several values in the 3 channels, the problem was that, in Figure 5 for example, in the pupil part when it is closer to the sclera, colours start to be

mixed and there are not clearly dark creating confusion with the vessels.

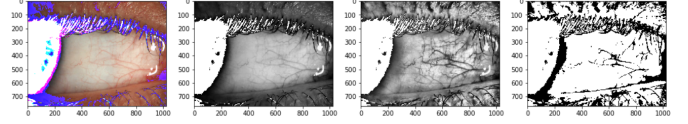


Fig. 5. Removing black parts

C. Color spaces analyzed

Apart from RGB colourspace, HSV (Figure 6) and Lab (Figure 7) spaces were also evaluated to check if the contrast between vessels and white areas where higher. But none of them was better than the original one.

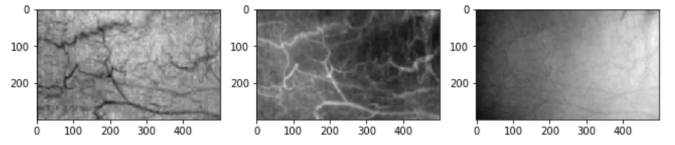


Fig. 6. HSV colorspace

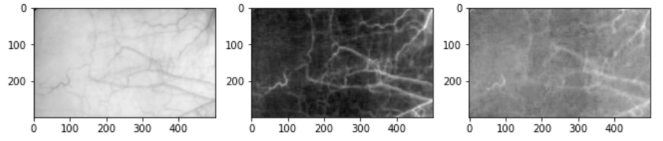


Fig. 7. Lab colorspace

D. Gamma correction

In order to improve the luminance of images where dark regions where present, a Gamma correction function was applied. However, after the process carried out and explained in the methodology, the results were pretty similar, so, basically, was taken out. A Gamma correction might be seen in Figure 8.

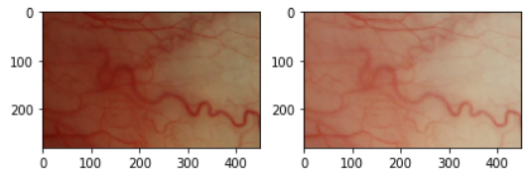


Fig. 8. Original cropped image and gamma applied

E. Morphological operations

Erosion transformation was also tested, see Figure 9, the main idea was to "clean" small vessels of the region after applying a thresholding method. It was discarded because results were not optimal.

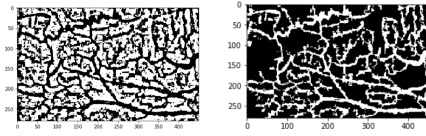


Fig. 9. Thresholding and erosion

F. Thresholding, Adaptive thresholding and average

Due to different lighting conditions in different areas of the images, adaptive thresholding was tested. With adaptive thresholding, the algorithm determines the threshold for a pixel based on a small region around it. After checking some configurations the approach was discarded due to the loss of vessel information. Also, a combination of thresholding and adaptive thresholding, applying the average between them for each pixel, was evaluated in order to maintain the maximum of vessels information discovered by thresholding avoiding the loss of adaptive thresholding but reducing the errors occasioned by the thresholding algorithm. Finally, this approach was also discarded due to poor results in some images.

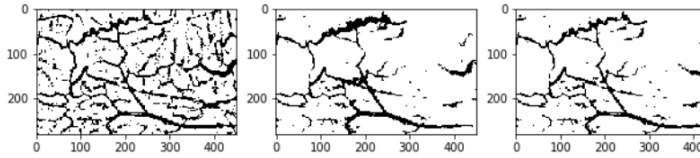


Fig. 10. Thresholding, adaptive thresholding and the average between them

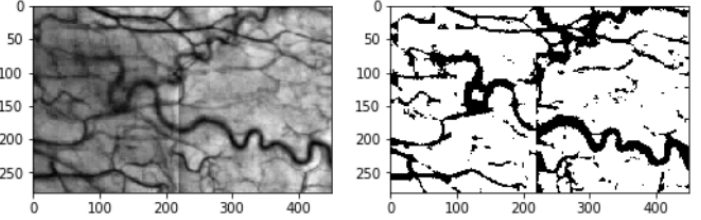


Fig. 11. Splitted image and thresholding independently

G. Image partition

As in the case explained before, due to different lighting conditions in different areas of the images a different approach was tried in order to avoid this. The idea was to split the input image into two halves; preprocessing was then done for each half independently: CLAHE for each part with the same parameters and, for the segmentation, Global thresholding technique was used, using as threshold the median of each half from the resultant values of the Equalization part. Actually, the results were very good, after defining the threshold for the classification, all the images were assigned in the correct ranges. But, it was finally discarded because images with high redness in the bulbar conjunctiva were having a quite similar percentage or value of occupancy than images with shadow regions.